

Blockchain-Based E-Commerce Platform with Integrated Escrow for the Kenyan Market

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Declaration

This proposal/research project is my original work and has not been presented for a degree in any other University.

Signature: _____  _____ Date: 27TH October, 2024

This proposal/research project has been submitted for examination with my approval as University Supervisor.

Supervisor: Dr. Dennis Kaburu

Signature: _____ Date: _____

Abstract

E-commerce and freelance marketplaces in Kenya have transformed how people buy, sell, and work, especially through social media platforms. However, trust issues, fraud, and lack of buyer protection remain significant challenges. This project aims to create a block chain-based e-commerce platform designed to address these problems. By using a secure escrow system, funds are held safely until the buyer confirms delivery, offering peace of mind for both buyers and sellers. The platform will also include decentralized dispute resolution and integrate social media verification to build trust and credibility for vendors. Developed using the Agile methodology and powered by the Polygon block chain, the solution is tailored to Kenya's unique market needs. This project is about more than just technology—it's about empowering people to trade confidently and transparently in an increasingly digital world.

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Chapter 1: Introduction

1.1. Background

E-commerce and freelance has transformed the way individuals and businesses conduct transactions globally, allowing purchases to happen across vast distances without the need for physical store presence. However, the rapid growth of social media platforms like TikTok, Instagram, and Facebook has brought unique challenges, particularly in markets where sellers lack physical stores, as is common in Kenya. The reliance on social platforms for product visibility also increases trust issues due to fraud risks and the difficulty in ensuring product authenticity and quality. The Kenyan market requires a secure, transparent system that allows buyers and sellers to transact with confidence, irrespective of a physical presence.

1.2. Project Overview

This project proposes a block chain-based e-commerce platform with an integrated escrow system that caters to both physical products and freelance services. It will create a virtual storefront for vendors and service providers without physical locations, securing payments through a block chain escrow until the product or service is confirmed as delivered. Using block chain, the platform ensures transparency, immutability, and trust, addressing concerns with security and verification. A decentralized dispute resolution system will handle conflicts, fostering a reliable, community-driven platform that instils buyer confidence.

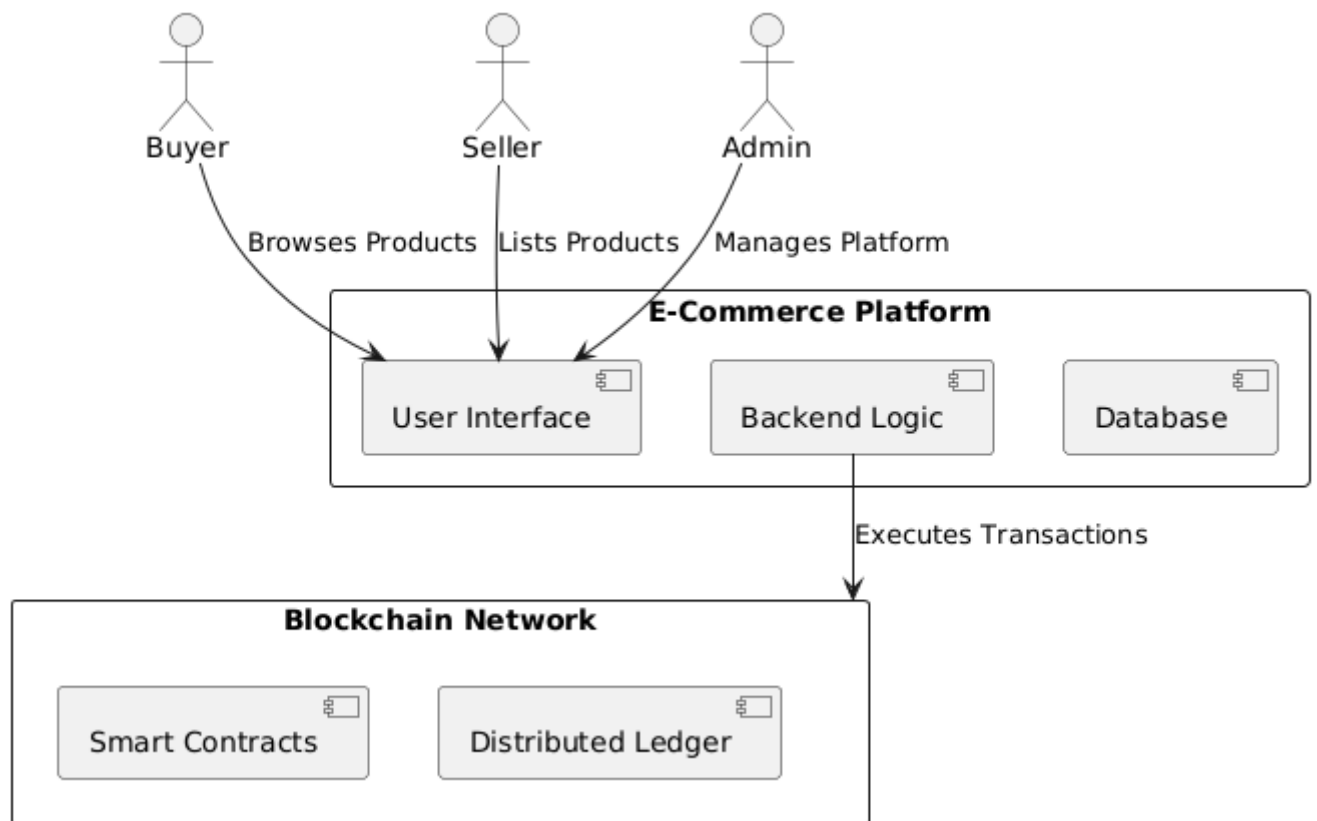


Figure 1. Context Diagram

1.3. Statement of the Problem

In Kenya, buyers face significant trust issues when purchasing from vendors on social media platforms, with frequent cases of fraud, delayed deliveries, or subpar products. The lack of physical storefronts makes it challenging for vendors to establish credibility, while buyers struggle with limited protections when transacting. The need for a secure system that holds funds in escrow until buyer satisfaction is confirmed is critical to protect both parties. Existing solutions are either limited to high-cost or centralized services that don't fully align with local market dynamics or address the rising popularity of freelance services.

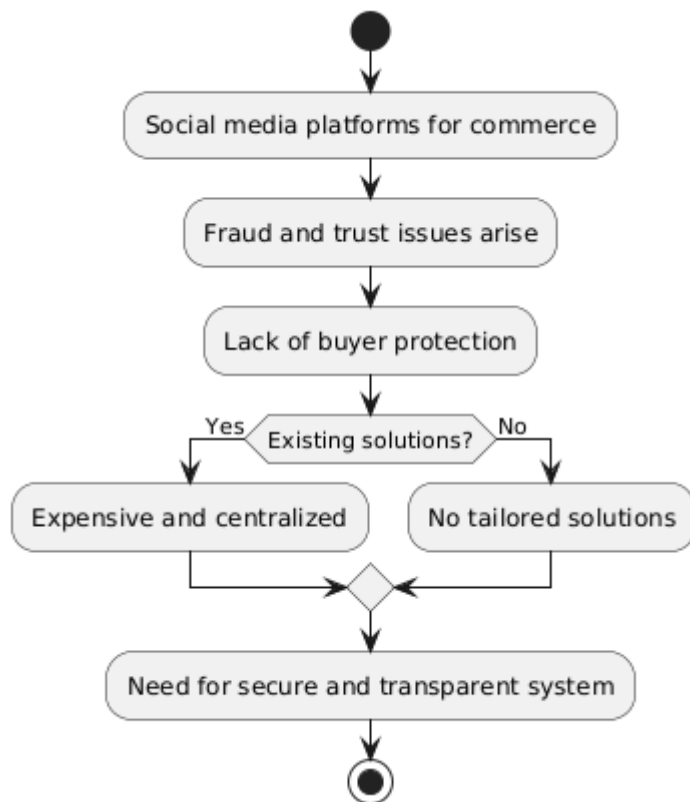


Figure 2. Problem Statement Flow

1.4. Proposed Solution

This project proposes a block chain-based platform that provides vendors with a digital storefront and offers escrow services, ensuring funds are only released upon delivery confirmation. Using decentralized smart contracts, it holds funds securely, adds a dispute resolution mechanism to manage conflicts, and supports social media integration for vendor verification. The platform will support both physical products and freelance services, aiming to build a comprehensive e-commerce solution that suits Kenyan market dynamics.

1.5. Objectives

- **General Objective:** To develop a block chain-based escrow e-commerce platform that supports both product and freelance service transactions for the Kenyan market.
- **Specific Objectives:**
 1. To conduct research on Digital Trust.
 2. Develop a secure digital block chain system.
 3. Document the development process of Block chain system.

1.6. Research Questions

1. What are the key ways block chain technology can enhance security?
2. What features are essential for a secure escrow system?
3. What methods improve vendor verification via social media?

1.7. Justification

The proposed platform will create a secure environment for online transactions in Kenya by leveraging block chain technology to reduce fraud risk, instilling trust in a market largely served by informal social media sellers. By supporting freelance services, the platform addresses a unique market need beyond physical products, expanding its applicability and appeal.

1.8. Proposed Research and System Methodologies

- **System Methodology:** The project will follow the agile methodology, allowing iterative development and testing of features.
- **Block chain Development:** Use Solidity for smart contract development and deploy contracts on the Polygon block chain.
- **Data Collection:** Research and surveys will be conducted to gather user needs and expectations.

1.9. Scope

The study focuses on developing an escrow platform tailored to Kenyan e-commerce for both products and freelance services. It will be limited to vendors without physical stores, and data collection will involve Kenyan buyers and sellers active on social media.

Chapter 2: Literature Review

2.1. Introduction

This literature review examines block chain's role in enhancing e-commerce security, with a particular focus on escrow systems and decentralized dispute resolution. It also explores social media's influence on buyer trust and reviews existing e-commerce solutions, identifying gaps addressed by the proposed platform.

2.2. Theoretical Review

Key concepts such as block chain, escrow, and decentralized arbitration will be examined. Block chain provides a decentralized, immutable ledger ideal for secure transactions, while escrow systems ensure funds are held until contractual obligations are met. Decentralized dispute resolution (e.g., Kleros) can handle conflicts transparently, building user trust without needing a central authority.

2.2.1. Key benefits of block chain in e-commerce

Block chain technology enhances e-commerce by providing:

- **Security:** Transactions are encrypted and stored on a decentralized ledger, reducing fraud and unauthorized alterations.
- **Transparency:** All participants can access transaction records, ensuring accountability.
- **Trust:** Smart contracts automatically execute agreements, eliminating the need for intermediaries.
- **Immutability:** Once recorded, transaction data cannot be altered, ensuring consistency and reliability.

2.2.2. Essential features for a secure escrow system

A secure block chain-based escrow system requires:

- **Smart Contracts:** Automate the holding and release of funds based on predefined conditions.
- **Dispute Resolution Mechanism:** Use decentralized arbitration to resolve conflicts.
- **Transparency:** Provide users with real-time updates on transaction progress.
- **Accessibility:** Ensure the system is user-friendly and supports various payment methods.
- **Scalability:** Handle increasing transaction volumes without compromising performance.

2.2.3. What methods improve vendor verification via social media?

Social media integration enhances vendor verification by:

- **Profile Linking:** Connecting vendor profiles to verified social media accounts for authenticity.
- **Reputation Scores:** Leveraging social media activity, reviews, and interactions to build credibility.
- **Community Feedback:** Using social media ratings and recommendations to improve trust.
- **API Integration:** Automating the retrieval of social media verification data for seamless user experience.

Global Trends

Block chain Adoption in E-Commerce: Worldwide, e-commerce platforms are increasingly adopting block chain to streamline operations, reduce fraud, and enhance transparency. Block chain's decentralized ledger system ensures that all transactions are immutable and transparent, fostering trust between buyers and sellers.

Integration with Emerging Technologies: The convergence of block chain with technologies like artificial intelligence (AI) is creating more robust e-commerce ecosystems. For instance, AI can analyse block chain data to provide personalized shopping experiences, while block chain ensures the security and authenticity of transactions.

Regional Trends in Kenya:

E-Commerce Growth: Kenya stands as the third-largest e-commerce market in Africa, with over 22.7 million internet users and a 40.8% internet penetration rate. This growth is driven by a young, tech-savvy population and widespread mobile technology adoption.

Block chain Adoption: Kenya is at the forefront of block chain adoption in Africa, implementing block chain solutions to stimulate economic development. The country's commitment to digital innovation positions it as a leader in block chain integration within the region.

Regulatory Developments: The Kenyan government is actively exploring block chain technology to address challenges in sectors like governance, healthcare, and agriculture. This proactive approach indicates a supportive environment for block chain-based e-commerce platforms.

2.3. Case Study Review

OpenBazaar: Decentralized Marketplace

OpenBazaar is an open-source, decentralized e-commerce platform that enables direct transactions between buyers and sellers without intermediaries. Utilizing block chain technology, it offers a peer-to-peer network where users can trade goods and services with greater privacy and reduced fees. The platform employs smart contracts to facilitate transactions, ensuring that funds are held in escrow until both parties fulfil their obligations. This approach enhances trust and security in online trading.

Upwork: Freelance Services with Escrow Payments

Upwork is a leading freelance marketplace that connects clients with freelancers across various industries. To ensure secure transactions, Upwork implements an escrow payment system where clients deposit funds into an escrow account before work begins. Once the freelancer completes the task to the client's satisfaction, the funds are released. This system mitigates risks for both parties, fostering trust and encouraging successful collaborations.

Walmart and IBM: Food Supply Chain Traceability

Walmart, in collaboration with IBM, has implemented a block chain-based system to enhance transparency and traceability in its food supply chain. By recording transactions on a block chain, Walmart can track the journey of food products from farms to stores, ensuring quality and safety. This system allows for rapid responses to contamination issues and builds consumer trust by providing transparent product information.

LVMH's AURA: Luxury Goods Authentication

LVMH, the parent company of brands like Louis Vuitton, has developed the AURA block chain platform to combat counterfeiting and verify the authenticity of luxury goods. By registering products on a block chain, AURA provides customers with a digital certificate of authenticity, detailing a product's history and proof of origin. This initiative enhances brand integrity and consumer confidence in the authenticity of their purchases.

Kleros: Decentralized Dispute Resolution

Kleros is a block chain-based platform offering decentralized dispute resolution services. It utilizes smart contracts and crowdsourced jurors to adjudicate disputes in various sectors, including e-commerce. By providing an impartial and transparent resolution process, Kleros enhances trust in online transactions and reduces reliance on traditional legal systems.

Key Successes and Limitations

Walmart's implementation of block chain technology in its food supply chain has led to significant improvements in traceability and safety. By collaborating with IBM to develop a system based on hyper ledger Fabric, Walmart reduced the time required to trace the origin of produce from seven days to just 2.2 seconds. This enhanced capability allows for rapid identification and removal of contaminated products, thereby minimizing the impact of foodborne illnesses. Additionally, the system's transparency fosters greater trust among consumers and suppliers.

However, the initiative also encountered challenges. Integrating block chain into a complex, global supply chain required significant coordination among diverse stakeholders, including farmers, distributors, and retailers. Ensuring data accuracy and consistency across the network was essential, as the reliability of the block chain depends on the integrity of the input data. Moreover, the adoption of new technology necessitated substantial investment in infrastructure and training. Despite these hurdles, Walmart's experience demonstrates the potential of block chain to enhance supply chain transparency and efficiency, serving as a model for other organizations considering similar implementations.

2.4. Integration and Architecture

Integrating block chain technology into an e-commerce platform with an escrow system involves a multi-layered architecture designed to ensure security, transparency, and efficiency. The proposed system comprises the following key components:

- **User Interface Layer:** This is the front-end component where buyers and sellers interact with the platform. It includes features such as product listings, shopping carts, payment options, and user profiles. The interface should be intuitive and accessible, catering to users with varying levels of technical expertise.

- **Application Layer:** Serving as the intermediary between the user interface and the block chain network, this layer manages business logic, processes user requests, and facilitates communication with the block chain. It handles tasks such as order processing, payment initiation, and status tracking.
- **Blockchain Layer:** At the core of the system, this layer utilizes block chain technology to record transactions immutably. Smart contracts deployed on the block chain automate the escrow process, holding funds until predefined conditions—such as delivery confirmation—are met. This ensures that transactions are secure and self-executing without the need for intermediaries.
- **Decentralized Storage:** To manage product information and user data securely, the system employs decentralized storage solutions like the Interplanetary File System (IPFS). This approach ensures data integrity and availability while reducing reliance on centralized servers
- **Social Media Integration Module:** This component interfaces with social media platforms to verify vendor identities and enhance trust. By integrating with APIs from platforms such as Facebook, Instagram, and TikTok, the system can authenticate vendors' social media profiles, providing buyers with additional assurance of the sellers' credibility.
- **Dispute Resolution Mechanism:** Incorporating decentralized dispute resolution protocols, such as Kleros, this module facilitates fair and transparent conflict resolution between parties. It enables community-driven arbitration, ensuring that disputes are settled impartially without centralized control.

2.5. Summary

In developing a block chain-based e-commerce platform with integrated escrow for the Kenyan market, I have examined various case studies to understand the successes and limitations of similar implementations. Platforms like OpenBazaar have demonstrated the potential of decentralized marketplaces by utilizing block

chain for anonymity and trust, employing an escrow system with moderators to resolve disputes between peers. However, OpenBazaar's lower volume and variety of items compared to established platforms highlight the challenge of attracting a diverse user base.

Additionally, the Kleros platform offers decentralized arbitration services, providing fast and affordable justice for e-commerce disputes. While Kleros showcases the effectiveness of decentralized dispute resolution, it also underscores the necessity for user education and acceptance of such novel systems.

Drawing from these insights, I plan to develop a platform that leverages blockchain technology to establish trust and transparency in transactions. By integrating a smart contract-based escrow system, funds will be securely held until the buyer confirms satisfactory delivery, thereby protecting both parties. To address potential disputes, I will implement a decentralized resolution mechanism inspired by models like Kleros, ensuring fair and efficient outcomes. Recognizing the importance of user trust, the platform will incorporate social media verification to enhance vendor credibility, catering specifically to the dynamics of the Kenyan market.

This approach aims to mitigate the limitations observed in other platforms by focusing on user engagement and trust-building measures, ultimately fostering a secure and reliable e-commerce environment.

2.6. Research Gaps

Current solutions do not fully cater to Kenyan market needs, particularly for small vendors and freelancers. Additionally, most platforms lack effective decentralized dispute resolution and social media verification, which this project aims to address.

References

1. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System.
2. Buterin, V. (2013). Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform.
3. Szabo, N. (1997). The idea of smart contracts.
4. Katina, M. (2018). Block chain and Trust: A Study of Block Chain's Impact on Trust in E-commerce Transactions. *Journal of E-commerce*, 12(3), 42-55.
5. Lee, S., & Lee, H. (2017). The Impact of Social Media on Online Trust in E-commerce. *International Journal of E-commerce Research*, 23(5), 67-85.
6. Wang, J., & Xu, R. (2021). Block chain Applications in Digital E-commerce: Security and Privacy Concerns. *Computing and Security*, 15(6), 91-108.
7. Gans, J. S., & Halaburda, H. (2015). Some Economics of Private Digital Currency. *Journal of Economic Perspectives*, 29(2), 127-150.
8. Kleros, F. (2020). Decentralized Justice: An Overview of Decentralized Dispute Resolution. Retrieved from <https://kleros.io/>
9. Turban, E., King, D., Lee, J., Liang, T., & Turban, D. C. (2015). Electronic Commerce: A Managerial and Social Networks Perspective. Springer.
10. Tapscott, D., & Tapscott, A. (2016). Block chain Revolution: How the Technology behind Bitcoin and Other Cryptocurrencies is changing the World. Penguin.

Appendices

Appendix A: Project Resources

- **Hardware:** Desktop/Laptop with sufficient computational power.
- **Software:** Visual Studio Code, Solidity, Metamask, IPFS, Polygon block chain network.
- **Libraries:** Web3.js, Truffle Suite, Mocha for testing.

Appendix B: Project Budget

Item	Description	Cost (USD)
Development Software	IDE, Truffle Suite	\$50
Block chain Fees	Gas fees on Polygon	\$100
Server Hosting	Web hosting for the platform	\$200
Miscellaneous	Surveys and Interviews	\$50
Total		\$400

Appendix C: Project Schedule

Task	Start Date
Literature Review	01/01/2024
System Design	16/01/2024
Smart Contract Development	01/02/2024
Testing and Evaluation	16/02/2024
Data Collection and Analysis	01/03/2024
Final Documentation	16/03/2024

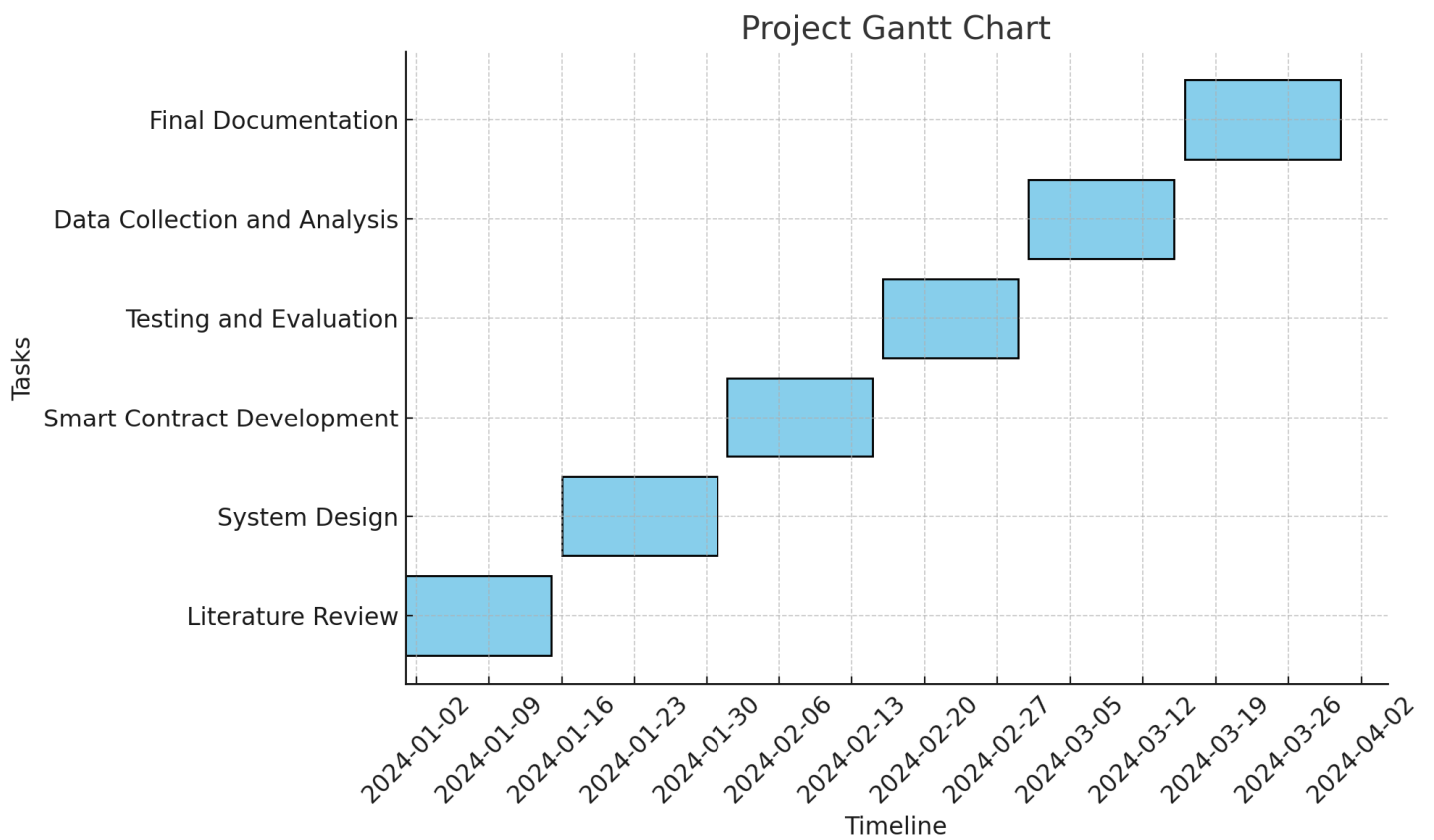


Figure 3. Gantt chart

PERT Chart for Blockchain-Based E-Commerce Project

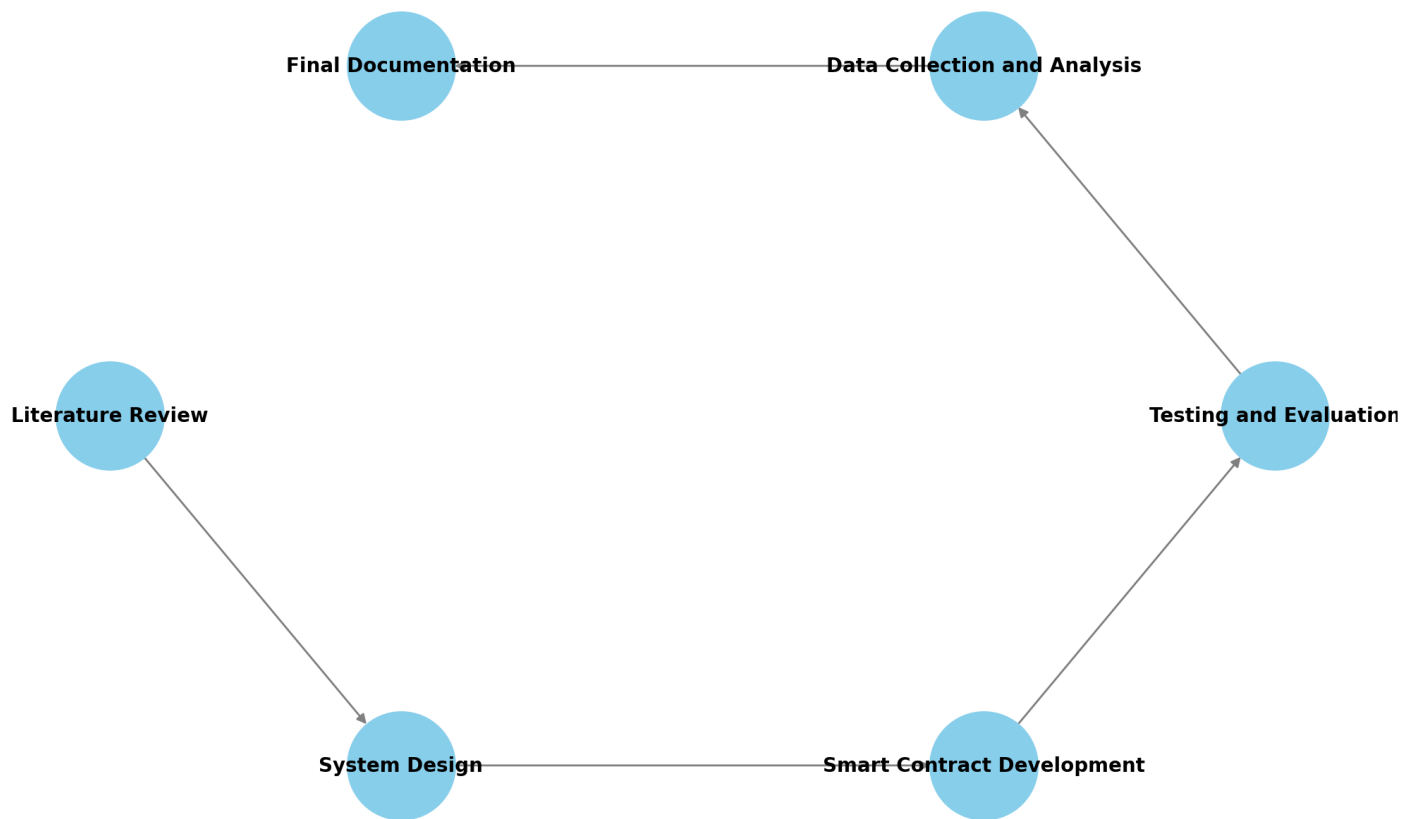


Figure 4. PERT Chart